

Technical Appendix: Higher Criticism for Adaptive Retrieval

Anonymous submission

Abstract

This supplementary document collects material referenced in the main paper but held out for space: the project roadmap, the HC-RAG pipeline diagram, the full comparison-methods table, the CrossEntityQA construction, and the null-distribution uniformity diagnostics.

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1 Project Roadmap

Wk	Milestone / Outcome
4	Foundations. Implemented the HC module (p -values from a per-query null, γ guardrail) and the indexing stack on the CRAG (Meta AI 2024) corpus (recursive chunking, 384/50 overlap; OpenRouter LLM interface). <i>Direction change:</i> a k -sweep with <code>all-mpnet-base-v2</code> favoured $k=10$, but BGE (Xiao et al. 2023) gave a clear recall lift on a 1K-doc subset, so BGE became the default embedder.
5	Survey of adaptive retrievers. Catalogued competing approaches (Reciprocal Rank Fusion, CAR (Xu et al. 2025), uncertainty filtering) to position HC as a global, distribution-aware test, and fixed the comparison set (Bonferroni (Dunn 1961), Benjamini–Hochberg (Benjamini and Hochberg 1995), gap-based Adaptive-K (Taguchi, Maekawa, and Bhutani 2025)). First HC vs. Top-K runs.
6	Long-context regime: HoloBench. First HC vs. gap-based Adaptive-K on HoloBench (Maekawa, Iso, and Bhutani 2024). Judge scores tracked closely; <i>iteration:</i> diagnosed as a generation/judge bottleneck on aggregation, not a retrieval tie, which motivated stronger retrieval-only metrics.
7	Multi-hop regime + dataset-aware γ. Repeated on HotpotQA (Yang et al. 2018). <i>Change in direction:</i> a stark dense-vs-sparse asymmetry meant the same γ behaved very differently, so γ was treated as dataset-aware. Began Adaptive-RAG (Jeong et al. 2024) and FlashRAG (Jin et al. 2025) integration.
8	Pipeline integrations + FiQA. Plugged HC into Adaptive-RAG as a fourth strategy; on HotpotQA it beat single-step ONE-R and lost to multi-step IRCOT, as expected. Registered HC and Berk–Jones (Berk and Jones 1979) as FlashRAG post-processors on BEIR SciFact (Thakur et al. 2021; Wadden et al. 2020), FiQA (Thakur et al. 2021), and MS MARCO. <i>Iteration:</i> FiQA was the first single-hop negative result and triggered the position study behind the scattered-relevance analysis.
9	CrossEntityQA construction. <i>Change in direction:</i> rather than keep tuning on off-the-shelf data, we built a sparse-relevance benchmark to surface HC’s target regime (SPARQL Wikidata clusters $\rightarrow$ Wikipedia passages $\rightarrow$ LLM queries answerable only by aggregating several passages; $\sim 2\%$ relevance, controllable K).
10	Negative result: CrossEntityQA + BM25. HC <i>lost</i> to fixed Top-5 under BM25, whose score magnitude encodes lexical match, not the categorical constraints these queries impose. <i>Change in direction:</i> dropped BM25 for CrossEntityQA and committed to a dense retriever.
11	Final breadth head-to-head. With the dense pipeline HC won on passage F1 on CrossEntityQA and was competitive with single-step retrieval on MuSiQue (Trivedi et al. 2022), closing the breadth picture and motivating the controlled Amazon study.

2 HC-RAG Pipeline

Figure 1 shows the full offline/online pipeline; every pipeline in the main paper differs only in the HC-cutoff box.

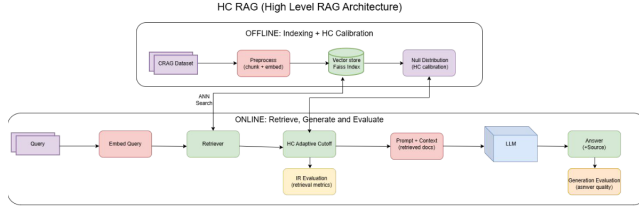


Figure 1: The HC-RAG pipeline. *Offline*: chunk and embed the corpus, build a FAISS (Johnson, Douze, and Jégou 2019) index, and calibrate the null. *Online*: embed the query, retrieve a candidate pool, apply the HC cutoff, and pass the selected passages to the LLM; retrieval and generation are both evaluated.

3 Comparison Methods

Table 1 lists the selection rules compared in the study. “Concept” is the rule that turns a candidate pool into a document set; the statistical methods share HC’s p -value machinery and differ only in the decision rule.

Method	Concept	Reference
Fixed Top- k	rank cutoff	—
Adaptive-K	largest score gap	(Taguchi,
Adaptive-RAG	complexity routing	(Jeong et al.
FlashRAG post-proc.	cutoff plug-in	(Jin et al.
Full-context	all chunks to the LLM	—
Self-route	retrieve + routing prompt	—
HC (ours)	Higher Criticism statistic	(Donoho and Jin 2004)
Berk-Jones	goodness-of-fit statistic	(Berk and Jones 1979)
Bonferroni	FWER threshold	(Dunn 1961)
Benjamini-Hochberg	FDR step-up	(Benjamini and Hochberg 1995)
CAR	cluster cutoff	(Xu et al. 2025)
Per-query recal. (ours)	rescale z -scores to the global null	—

Table 1: Methods compared. All share HC’s p -value machinery except the fixed and cluster rules.

4 CrossEntityQA Construction

Figure 2 shows the CrossEntityQA construction pipeline. It guarantees a controllable per-query relevance count ($K=2-20$) and a sparse corpus (about 2% relevant), the two properties that make the benchmark a stress test for adaptive- k retrieval.

5 Null-Distribution Diagnostics

This appendix verifies that the p -values are uniform under the null. On BEIR FiQA, under the *empirical* null the non-relevant p -values are uniform (KS $p=0.13$; 5.0% below 0.05), whereas a *parametric normal* null is not (KS $p=0.000$, an excess spike near zero), which is why we keep the empirical

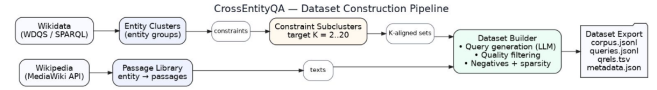


Figure 2: CrossEntityQA construction. Wikidata entity groups (WDQS/SPARQL) are split into constraint subclusters with a target relevance count $K=2-20$; a Wikipedia passage library supplies the texts; the builder generates LLM queries, filters for quality, and controls negatives and sparsity, exporting corpus, queries, and qrels.

null; relevant documents carry systematically small p -values (Cohen’s $d=-1.33$), and the QQ plot of empirical p -values sits on the uniform diagonal (Figure 3).

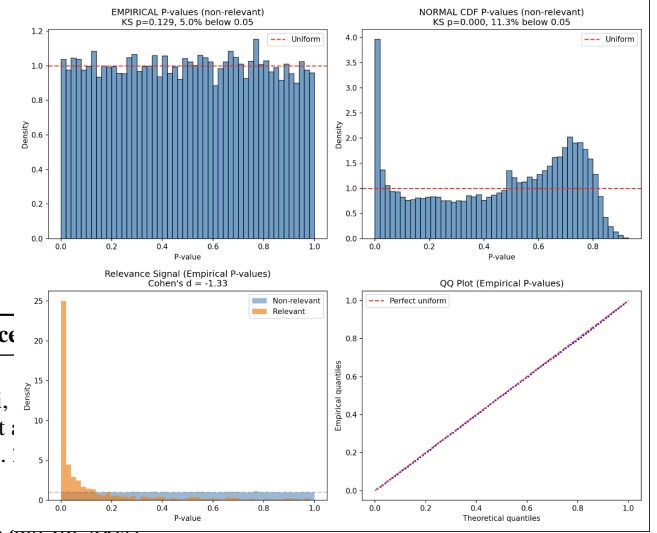


Figure 3: p -value uniformity diagnostics under the null (BEIR FiQA). The empirical null is uniform (top-left; QQ plot bottom-right) while a parametric normal null is not (top-right); relevant documents separate in p -value space (bottom-left).

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